

Lab 05

→ Objectives

The objective of this lab is to introduce you to UNIX file system and properties of files.

→ Description

We'll start with the *stat* and *lstat* functions. Both functions return a structure called *stat*, and members of *stat* structure provide information about the file or directory which was provided as the argument to these functions.

Example 1:

- ☒ Now let's write program to print the type of file
 - For any given command-line argument
- ☒ Use *lstat* function ...
 - Similar to *stat* function, except that when there is a symbolic link the *lstat* function provides details about the link itself not the file the link points to.

Example 2:

Till now we talked about how filesystem store files and directories information and access details. Now it's time to learn how to open and read the directories and traverse the file system. To achieve this task, you need to learn about three functions:

- 1) *opendir* – this function will allow us to open a directory with the given path
- 2) *readdir* – this function will read what's inside the directory
- 3) *closedir* – this will close the open directory

- ☒ Write a simple program to open, read, and close a directory

NOTE: We can use the read/write system calls from Lab04 to write the *stat* structure and read the *stat* structure. Data is written as a binary file.

→ Homework

Extend the *readdir* program ...

- ☒ traverse the file hierarchy recursively
- ☒ list all the sub directories and the files in these sub directories
- ☒ include README.txt ...

NOTE: have to modify the given readdir.c or program to use a function to perform the file traversal and then invoke it recursively.